

OSSP

Practical Week -2

Roll No – 2520090085

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Task 1-:

Develop a C program that uses the system calls `open()`, `read()`, `write()`, and `close()` to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

```
GNU nano 8.7.1 file_copy.c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <fcntl.h>

#define BUFFER_SIZE 1024

int main(int argc, char *argv[]) {
    if (argc != 3) {
        write(STDERR_FILENO, "Usage: ./file_copy <source_file> <dest_file>\n", 45);
        exit(EXIT_FAILURE);
    }

    int src_fd = open(argv[1], O_RDONLY);
    if (src_fd < 0) {
        perror("Error opening source file");
        exit(EXIT_FAILURE);
    }

    int dest_fd = open(argv[2], O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (dest_fd < 0) {
        perror("Error opening/creating destination file");
        close(src_fd);
        exit(EXIT_FAILURE);
    }

    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    while ((bytes_read = read(src_fd, buffer, BUFFER_SIZE)) > 0) {
        bytes_written = write(dest_fd, buffer, bytes_read);
        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(src_fd);
            close(dest_fd);
            exit(EXIT_FAILURE);
        }
    }
}
```

```

        exit(EXIT_FAILURE);
    }

    char buffer[BUFFER_SIZE];
    ssize_t bytes_read, bytes_written;

    while ((bytes_read = read(src_fd, buffer, BUFFER_SIZE)) > 0) {
        bytes_written = write(dest_fd, buffer, bytes_read);
        if (bytes_written != bytes_read) {
            perror("Error writing to destination file");
            close(src_fd);
            close(dest_fd);
            exit(EXIT_FAILURE);
        }
    }

    if (bytes_read < 0) {
        perror("Error reading from source file");
    }

    close(src_fd);
    close(dest_fd);

    printf("File copied successfully.\n");
    return 0;
}

```

Output:-

```

nocturnal@nocturnal:~/OSSP_2520090085$ mkdir P2
nocturnal@nocturnal:~/OSSP_2520090085$ nano file_copy.c
nocturnal@nocturnal:~/OSSP_2520090085$ gcc file_copy.c -o file_copy
nocturnal@nocturnal:~/OSSP_2520090085$ echo "Hello from user space and kernel space!" > source.txt
nocturnal@nocturnal:~/OSSP_2520090085$ ./file_copy source.txt destination.txt
File copied successfully.
nocturnal@nocturnal:~/OSSP_2520090085$ cat destination.txt
Hello from user space and kernel space!

```

Task 2

Use the strace utility to trace all system calls generated by the command `cat sample.txt`. Analyze the sequence of system calls and identify the kernel services involved.

Output:-

[illegible]